

Savannah L. Ferretti

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EDUCATION

Ph.D. in Earth System Science, University of California, Irvine Jun 2026
M.S. in Earth System Science, University of California, Irvine Dec 2023
B.S. in Earth & Atmospheric Sciences, Cornell University Dec 2020

EXPERIENCE

Postdoctoral Scholar Aug–Oct 2026
Pritchard & Baldwin Groups, University of California, Irvine Irvine, CA

- Extended PhD research on South Asian monsoon rainfall variability and prepared work for publication

Graduate Student Researcher Sep 2021–Jun 2026
Pritchard & Baldwin Groups, University of California, Irvine Irvine, CA

- Used data-driven and physics-based approaches to identify & explain drivers of South Asian monsoon rainfall variability
- Developed an interpretable machine learning methodology that used neural networks & equation discovery to learn physically meaningful relationships from high-dimensional atmospheric data
- Built open-source Python pipelines on GitHub for cloud data access, large-array processing, uncertainty analysis, model evaluation, & visualization in HPC environments
- Communicated findings through peer-reviewed publications & conference presentations

Consulting Analyst Mar–Aug 2021
Higher Education Research Services, Huron Consulting Group New York, NY

- Built Excel-based financial planning & reporting tools for a research nonprofit
- Developed long-term resource forecasts & presented recommendations to executive stakeholders

Undergraduate Research Assistant Jun–Aug 2020
Commene Group, Lamont-Doherty Earth Observatory Palisades, NY

- Quantified changes in atmospheric methane concentrations during the NYC COVID-19 shutdown
- Applied meteorological analysis & transport modeling to attribute emissions to likely source regions

SKILLS

Python (PyTorch, PySR, Dask, Xarray, Cartopy, Matplotlib) • GitHub • High-Performance Computing
• NetCDF/HDF5/Zarr • Bash • Linux • Microsoft Excel ([Certified Specialist](#)) • $\text{\LaTeX}$ • WordPress

SELECTED PUBLICATIONS

S. L. Ferretti, J. Lin, S. Shamekh, J. W. Baldwin, M. S. Pritchard, & T. Beucler. (2026). [Data-driven integration kernels for interpretable nonlocal operator learning](#). In press at *Environ. Data Sci.*

S. L. Ferretti, M. S. Pritchard, F. Ahmed, L. Peng, & J. W. Baldwin. (2025). [Explaining South Asian monsoon rainfall seasonality using a metric of plume buoyancy](#). *Geophys. Res. Lett.*